NTOU Java Programming
Exercise 4

SOSEL
Service-Oriented Software Engineering Lab

Spring 2026

Exercise Goals

- Object Interactions
- Exception Handling
- 2-Dimensional Object Arrays

Lab Instructions1

- Create a class called *Complex* for performing arithmetic with complex numbers. Complex numbers have the form

  *realPart + imaginaryPart * i*

  - Use *double* variables *real* and *imaginary* to represent the private data of the class.
  - Provide a constructor that enables an object of this class to be initialized when it is declared. (以設定實部與虛部)
  - Provide a no-argument constructor with random non-negative double values (from 0 to 1).
    - Please use *java.security.SecureRandom* to generate random numbers.

Lab Instructions2

- Provide public methods that perform the following operations:
  - Add two Complex numbers
  - Subtract two Complex numbers
  - Multiply two Complex numbers
  - Return a string representation in the form a + bi, where a is the real part and b is the imaginary part. (顯示到小數點後兩位)
- References:
  - https://en.wikipedia.org/wiki/Complex_number
  - https://zh.wikibooks.org/zh-tw/複數

Lab Instructions3

- Based on the developed code, we can develop a Quantum Computing Simulator (量子計算模擬器) that processes quantum states represented as complex vectors.
  - Each quantum state is stored as a 1D array of complex numbers, and multiple quantum states form a 2D matrix.
- To analyze these quantum states, we need to implement a function that follows these steps:
  - Sum each row: Compute the sum of all complex numbers in each row. (對每一行的複數數值進行加總)
  - Multiply the row sums: Multiply all the summed results together to get a final complex number. (再將這些加總結果彼此相乘)

Lab Instructions4

- Accordingly, please develop another class *ComplexNumberCalculator*
  - Implement a method: *public static Complex calculate(Complex[][] matrix)*
  - Add exception handling to check for invalid inputs
    - If the input matrix is null, has zero rows, or contains null/empty rows, throw an IllegalArgumentException.
- Please execute the provided "ComplexTest".

Expected Results


```
a = 1.10 + 2.20i
b = 3.30 - 4.40i
a + b = 4.40 - 2.20i
a - b = -2.20 + 6.60i
a × b = 13.31 + 2.42i

c = 0.69 + 0.80i
d = 0.43 + 0.35i
c + d = 1.12 + 1.15i
c - d = 0.26 + 0.44i
c × d = 0.02 + 0.59i

Simulation result: -4.84 + 14.52i
java.lang.IllegalArgumentException: Input matrix
cannot be null or empty
```

Hint

- For the *add, subtract, and multiply* methods of class *Complex*, return a new Complex object with the results of the calculations.
  - You need *java.lang.Math*
- To print an object, please add a public method: *toString()* that returns a String.
  - You can refer to https://www.javatpoint.com/understanding-toString()-method
- Please be familiar with handling two-dimensional arrays.

Submission

- The naming should conform to the **CamelCase** style.
- "Package" is required: ntou.cs.java2026.
- Please submit .java files and .class files (upload them to TronClass).
- Code that fails to compile or execute is not accepted.